Distributed Trip-Level Fuel Consumption Forecasting

from raw vehicle telemetry to trip-level fuel consumption prediction

Big Data Final Project

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Project and Objective

Problem: Raw vehicle telemetry is sensor-level: one trip consists of many timestamped rows. It cannot be directly used for trip-level ML.

Objective: Build a distributed pipeline that converts raw telemetry into clean trip-level features, analyzes fuel-consumption patterns, trains forecasting models, and presents the results in Superset.

Why it matters: Fleet operators need to understand which trips and conditions lead to inefficient fuel consumption.

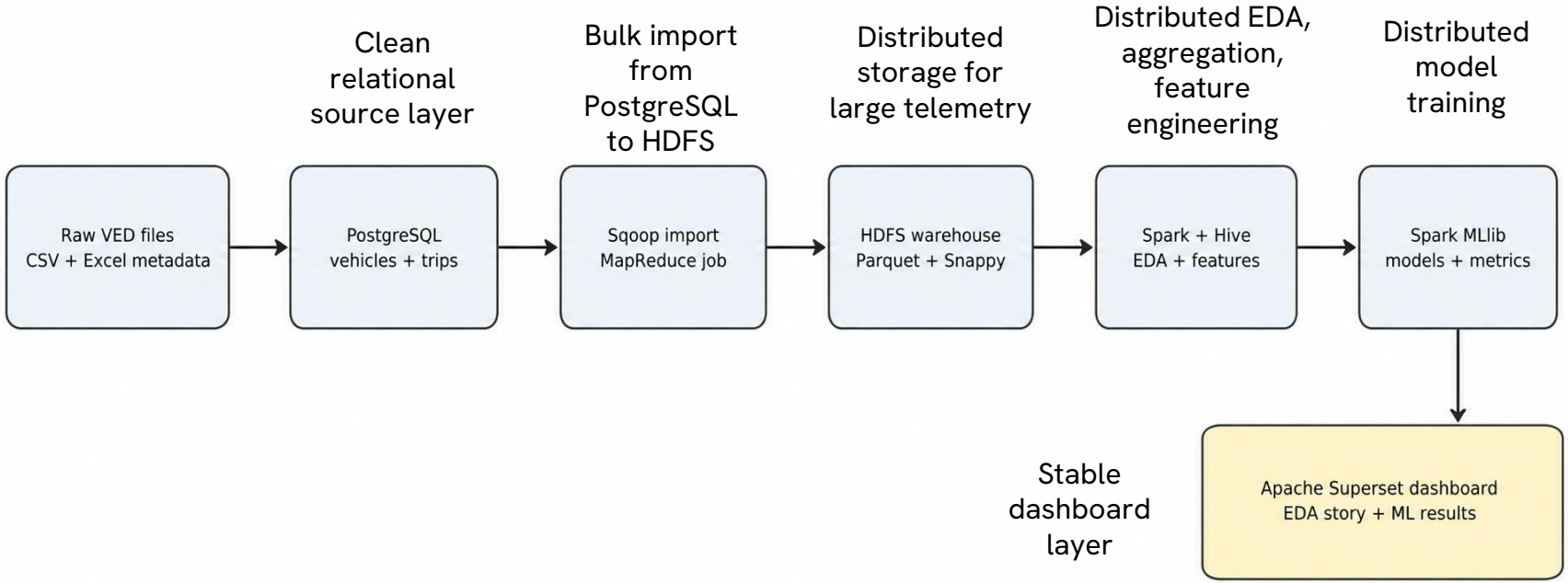
Dataset Characteristics

Metric	Value
Raw telemetry rows	22.4M
Vehicles	384
Trips before filtering	~32.5k
Trainable trips	~25.2k
Target	fuel_l_per_100km

Why distributed processing?

Because the raw telemetry table is large, sensor-level, and repeatedly processed for profiling, EDA, feature engineering, and ML

Pipeline Architecture



Stage 1: Data Collection and Ingestion

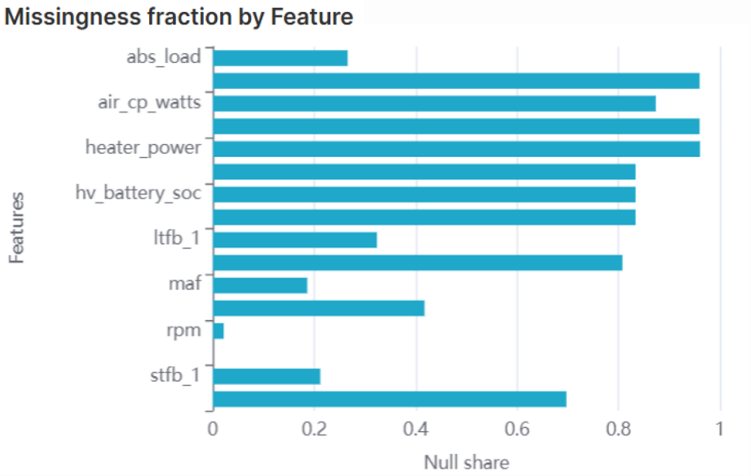
Stage 1. What we did:

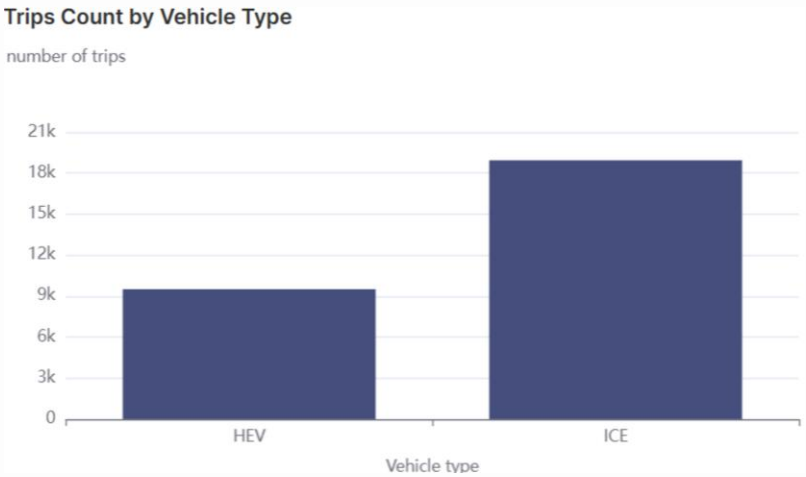
- Collected raw VED telemetry and vehicle metadata
- Created PostgreSQL tables: *vehicles* and *trips*
- Loaded source data into PostgreSQL
- Imported data to HDFS using Sqoop
- Stored warehouse data as Parquet + Snappy

Why?

- PostgreSQL gave us a structured relational source layer
- Sqoop was suitable for bulk transfer from PostgreSQL to HDFS
- Parquet + Snappy was chosen because the workload is read-heavy and column-oriented

Stage 2: Data Quality and Fleet Composition





Key problem:

- Direct *fuel_rate* is missing for about 96% of raw telemetry rows, so we cannot rely only on direct fuel measurements.
- Vehicle type imbalance must be considered.

From Sensor Rows to Trip-Level Features

What we did:

- Joined *trips* with *vehicles* by *vehid*.
- Estimated fuel rate using MAF when direct *fuel_rate* was missing.
- Aggregated sensor-level records into one row per (*vehid*, *tripid*).
- Created ML-ready trip-level features.

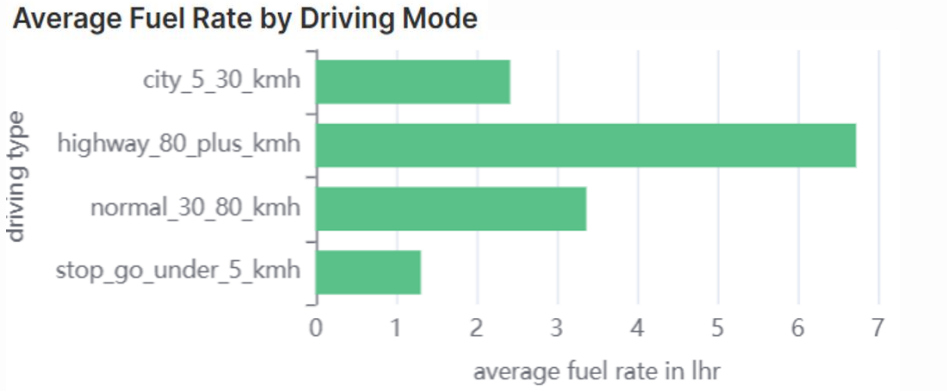
Why?

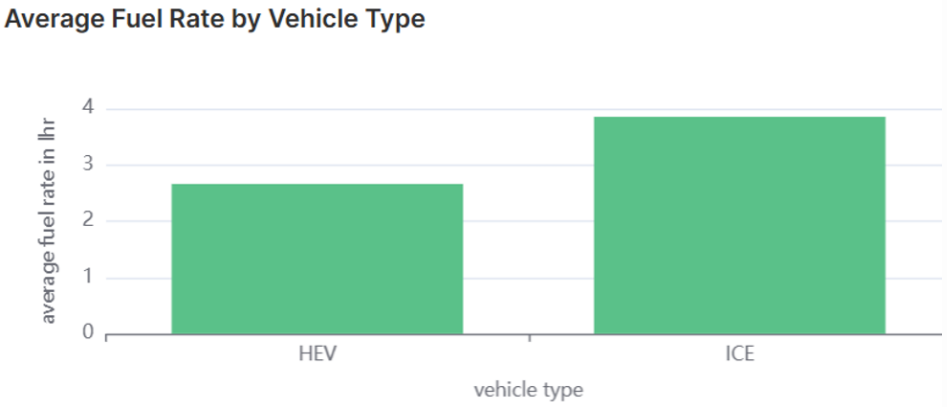
- ML target is trip-level fuel consumption, not timestamp-level sensor value.
- Vehicle metadata is needed because vehicle type, weight, engine displacement, and drivetrain affect fuel consumption.
- MAF approximation keeps more data usable despite sparse direct fuel measurements.

EDA: How this affects ML

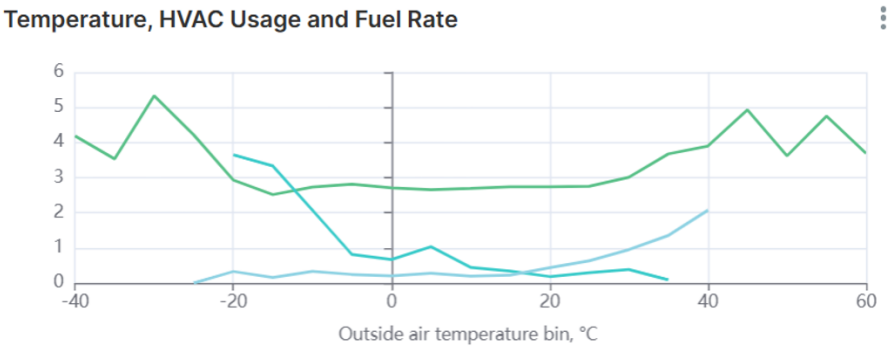
Findings:

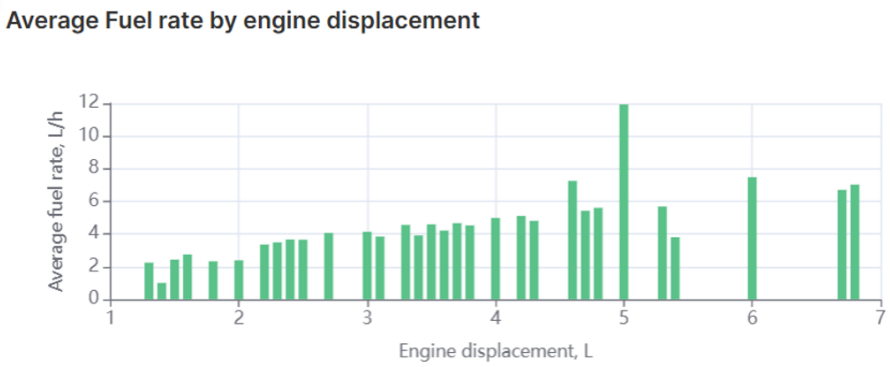
- Driving mode and vehicle type were kept as important explanatory features because fuel behavior differs across these groups.
- ICE vehicles consume more fuel on average than HEV vehicles.
- Driving mode and vehicle type are important ML features.





EDA Result: Feature Effects





Main observations:

- The relationship between fuel consumption and speed is nonlinear, so we tested tree-based models in addition to Linear Regression.
- Fuel rate increases with speed.
- Larger engines usually have higher fuel rate.
- Most fuel consumption values are in a realistic range.
- Extreme values should be filtered before ML.

Stage 3: Spark ML Modeling

ML task: predict *fuel_l_per_100km* for each trip.

Pipeline:

Clean trip-level features → outlier filtering → feature selection → train/test split →
→ Spark MLlib models → evaluation

Why these steps:

- Filtering removes unrealistic trips and target outliers.
- Mutual information was used because fuel consumption has nonlinear relationships with features.
- RMSE, MAE, and R^2 were used to evaluate both error size and explained variance.
- Linear Regression is a baseline. Random Forest and GBT were used because EDA showed nonlinear fuel behavior.

Model	RMSE	MAE	R2
Linear Regression	1.9562	1.0059	0.7804
RandomForestRegressor	1.3968	0.7564	0.8882
GBT Regressor	1.4696	0.7656	0.8762

Stage 4: Superset Dashboard

Dashboard structure:

- **Data Description** — explains raw data scale, schemas, and source tables.
- **EDA** — shows data quality and fuel-consumption patterns.
- **PDA** — presents ML results, best model, metrics, and selected features.

Why this structure:
It follows the project logic: first understand the data, then analyze patterns, then evaluate predictive modeling.

Challenges

Problem	Our decision	Why
Sparse fuel_rate	MAF-based approximation	Keeps more trips usable
Sensor-level rows	Trip-level aggregation	ML target is trip-level
Read-heavy analytics	Parquet + Snappy	Faster selected-column reads
Nonlinear fuel behavior	Tree-based models	Better fit for speed/mode interactions

Conclusions

Fuel consumption is strongly related to speed, driving mode, vehicle type, engine displacement, temperature, and HVAC usage

MAF-based approximation and trip-level aggregation made the data usable for analysis and forecasting

Missingness is the main data-quality limitation

Tree-based models performed better than a simple linear baseline

Thank you for attention!